

Architecting a Small Language Model (SLM)

FROM SCRATCH: SANSKRIT TO ENGLISH TRANSLATION OF THE BHAGAVAD GITA

Abstract: This document outlines the architectural blueprint for designing, training, and deploying a Small Language Model (SLM) specifically tailored for translating Sanskrit texts—with a prime focus on the Bhagavad Gita—into English. The architecture strictly adheres to a "from scratch" methodology, leveraging a custom, highly efficient Transformer model without relying on massive pre-trained weights. The design prioritizes computational efficiency, precise tokenization of ancient syntax, and context-aware philosophical translation.

1. Introduction & Objectives

While Large Language Models (LLMs) possess vast multilingual capabilities, they are computationally heavy and often hallucinate cultural or philosophical nuances inherent in ancient texts. A Small Language Model (SLM)—typically ranging from 50 Million to 150 Million parameters—offers a focused, domain-specific alternative.

The objective is to train a sequence-to-sequence (Seq2Seq) model from scratch that translates Sanskrit slokas (verses) to English, capturing the theological and philosophical accuracy of the Bhagavad Gita.

2. Dataset Strategy & Acquisition

Training an SLM from scratch requires a robust, high-quality parallel corpus. Since the Bhagavad Gita alone consists of only 700 verses (which is insufficient for from-scratch training), a two-stage data strategy is implemented using readily available datasets.

Phase 1: Base Language Acquisition (General Sanskrit-English)

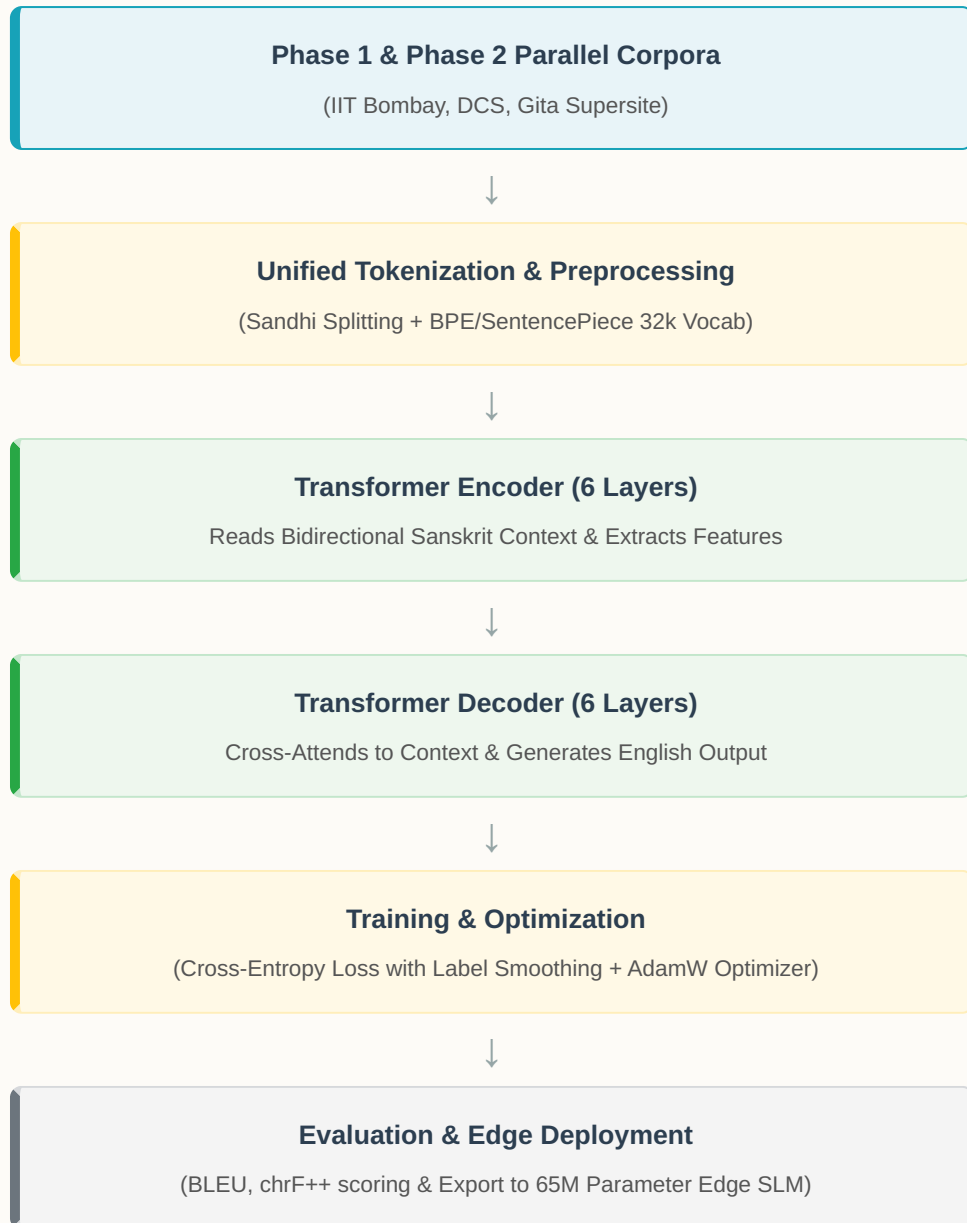
- **IIT Bombay English-Hindi/Sanskrit Corpus:** Portions of this established dataset containing Sanskrit-English alignments.
- **JNU Sanskrit Corpora:** Linguistic datasets containing grammatical structures, root words (Dhatus), and English equivalents.
- **Digital Corpus of Sanskrit (DCS):** Lexical resources mapping Sanskrit terms to English meanings.

Phase 2: Domain-Specific Fine-Tuning (Bhagavad Gita)

- **Gita Supersite (IIT Kanpur):** Parallel data containing Devanagari verses, IAST transliteration, and multiple English translations.
- **Hugging Face Datasets:** Existing repositories (e.g., `sanskrit_english_parallel`) containing structured JSON of Gita verses.

3. System Architecture & Flow

For a pure translation task at a small scale, a **Transformer Encoder-Decoder** architecture is vastly superior and more efficient than a Decoder-only (GPT-style) model. The Encoder reads the full Sanskrit verse to grasp the bidirectional context, while the Decoder generates the English text sequentially.



SLM Blueprint Details: "Gita-Translate-Small"

Total Parameters: ~65 Million (Optimized for edge deployment and fast inference)

COMPONENT	SPECIFICATION	RATIONALE
Model Type	Transformer Encoder-Decoder	Ideal for Seq2Seq tasks; captures bidirectional context of complex Sanskrit grammar.
Layers	6 Encoder / 6 Decoder	Sufficient depth for linguistic mapping without overfitting a small corpus.
Attention Heads	8	Allows the model to attend to distinct grammatical rules, prefixes, and suffixes simultaneously.
Embedding Dimension	512	Provides enough dimensional space for semantic meaning while keeping memory footprint low.
Feed-Forward Dim	2048	Standard expansion ratio (4x) for internal non-linear transformations.

4. Tokenization Strategy

Sanskrit is a highly agglutinative language where words are often fused together through *Sandhi* (phonetic junctions). A standard English tokenizer will fail to process this.

- **Algorithm:** Byte-Pair Encoding (BPE) or Unigram-based SentencePiece.
- **Vocabulary Size:** Custom vocabulary of 32,000 tokens (shared between Sanskrit and English).
- **Preprocessing:** The Sanskrit text is standardized to the International Alphabet of Sanskrit Transliteration (IAST) or preserved in Devanagari. Training a single tokenizer on a combined corpus ensures the model can bridge sub-word elements effectively.

5. Training Methodology (From Scratch)

Because the architecture contains no pre-trained weights, the training pipeline must be rigorously structured.

Objective Function

The model is trained using **Cross-Entropy Loss** with Label Smoothing (typically 0.1). Label smoothing prevents the model from becoming overly confident in its translations, which is crucial given that a single Sanskrit word can have multiple philosophical interpretations.

Optimization & Hardware Profile

An **AdamW** optimizer is used with a learning rate scheduler featuring a warm-up phase followed by a linear decay. Due to the 65M parameter constraint, this SLM can be trained from scratch on a single consumer-grade GPU (e.g., NVIDIA RTX 3090 or 4090) within a matter of days, making it highly accessible.

6. Evaluation Metrics

Translation quality is assessed using a blend of automated metrics and human-in-the-loop validation.

- **BLEU Score (Bilingual Evaluation Understudy):** Measures precise n-gram overlap between the model's output and the reference human translation.
- **chrF++:** A character n-gram metric that correlates highly with human judgment, especially useful for morphologically rich inputs like Sanskrit.
- **Semantic Preservation (Human Eval):** Automated metrics often fail to capture profound theological meaning. A subset of outputs must be reviewed for tone, philosophical accuracy, and readability.

7. Conclusion

By constraining the scope to a 65-million parameter Encoder-Decoder architecture, this SLM offers a highly targeted, computationally inexpensive solution for Sanskrit-to-English translation. Leveraging readily available public datasets for base training and fine-tuning specifically on the Bhagavad Gita ensures that the output is not just grammatically correct, but philosophically resonant. The absence of bloat—achieved by training from scratch—results in a fast, efficient model perfectly suited for academic, spiritual, and edge-device applications.